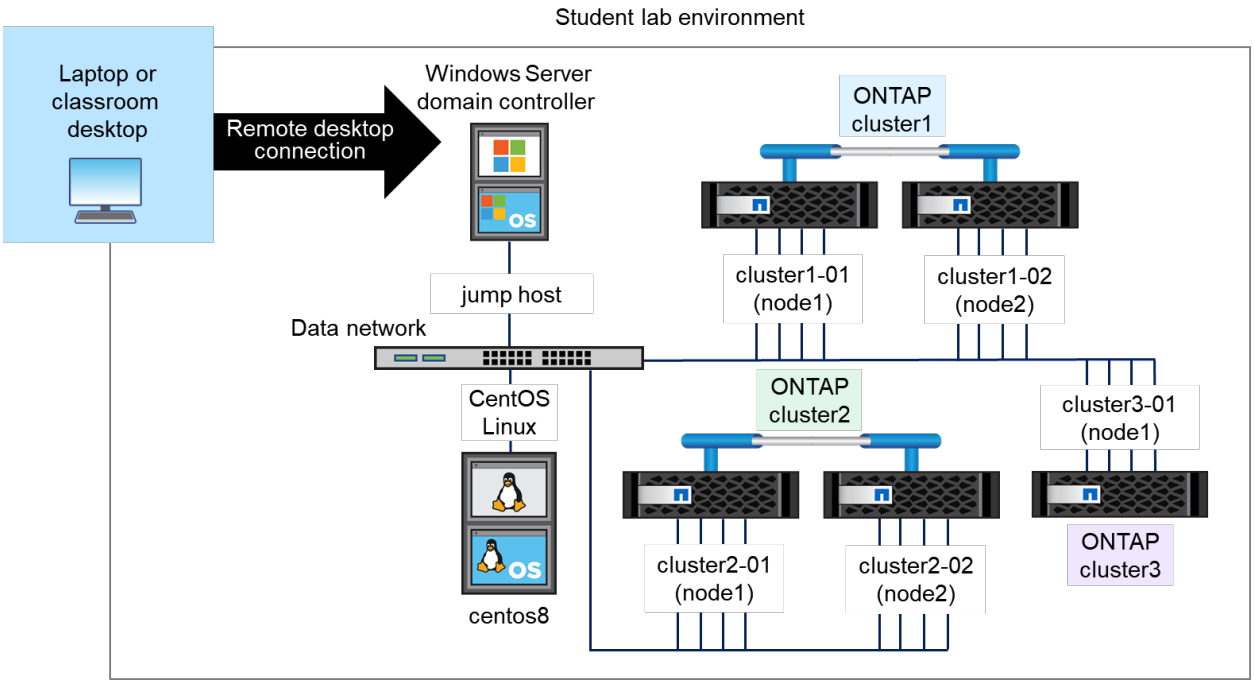
Module 1: SnapMirror Fundamentals

Exercise Equipment Diagram



System	Host Name	IP Addresses	User Name	Password
Windows Server 2022 Datacenter	JUMPHOST	192.168.0.5	DEMO\Administrator	Netapp1!
ONTAP cluster-management LIF (Cluster1)	Cluster1	192.168.0.101	admin (case-sensitive)	Netapp1!
node 1 (Cluster1)	cluster1-01-console	192.168.0.111	admin (case-sensitive)	Netapp1!
node 2 (Cluster1)	cluster1-02-console	192.168.0.112	admin (case-sensitive)	Netapp1!
ONTAP cluster-management LIF (Cluster2)	Cluster2	192.168.0 102	admin (case-sensitive)	Netapp1!
node 1 (Cluster2)	cluster2-01-console	192.168.0 113	admin (case-sensitive)	Netapp1!
node 2 (Cluster2)	cluster2-02-console	192.168.0.114	admin (case-sensitive)	Netapp1!
ONTAP cluster-management LIF (Cluster3)	Cluster3	192.168.0.103	admin (case-sensitive)	Netapp1!
node 1 (Cluster3)	cluster3-01-console	192.168.0.115	admin (case-sensitive)	Netapp1!
Linux Server	centos8	192.168.0.21	root	Netapp1!

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Exercise 1: Configuring Cluster Peering and Storage VM Peering

In this exercise, you use NetApp ONTAP System Manager to set up the prerequisite configurations for data protection between Cluster1 (the primary ONTAP cluster) and Cluster2 (the secondary ONTAP cluster).

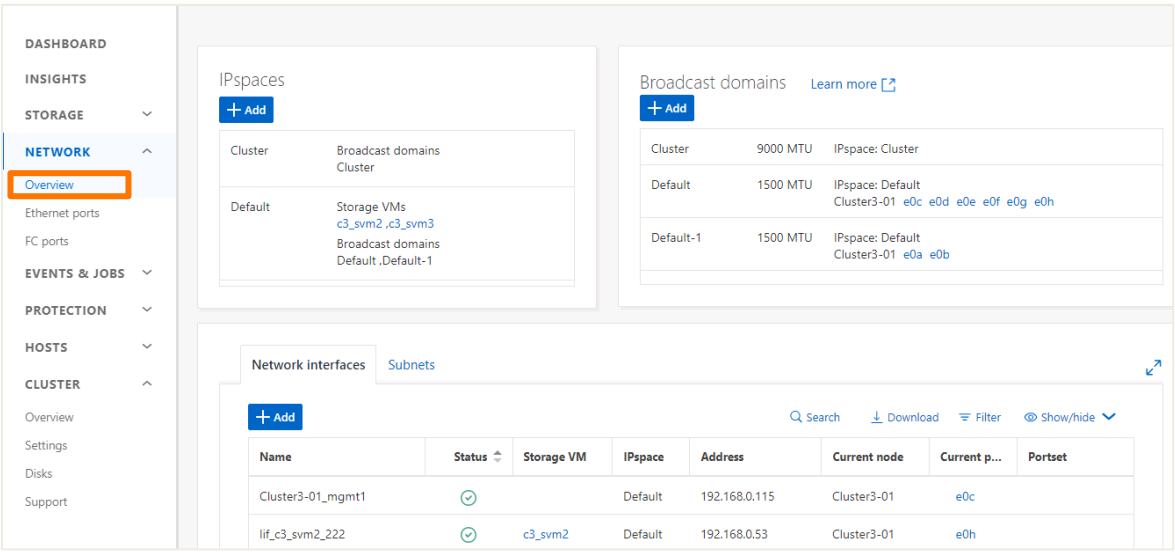
Objectives

This exercise focuses on enabling you to do the following:

- Create intercluster LIFs for both clusters
- Configure cluster peering
- Configure storage VM (storage virtual machine, also known as SVM) peering
- Review the exercise environment

Task 1: Create Intercluster LIFs on Cluster1

In this task, you implement the network connectivity to enable peering relationships between clusters. In the following steps, you create intercluster LIFs for Cluster1 and Cluster2.

Step	Action
1-1	When you use the ONTAP CLI to perform the exercises, you often switch between PuTTY sessions for each cluster. To help you avoid mistakes, exercise steps that require you to switch between clusters include the corresponding cluster label. Steps that occur on Cluster1 begin with a Cluster1 label. Steps that occur on Cluster2 begin with a Cluster2 label. Steps that occur on Cluster3 begin with a Cluster3 label.
1-2	In the System Manager navigation pane for Cluster1, select **Network > Overview**. 

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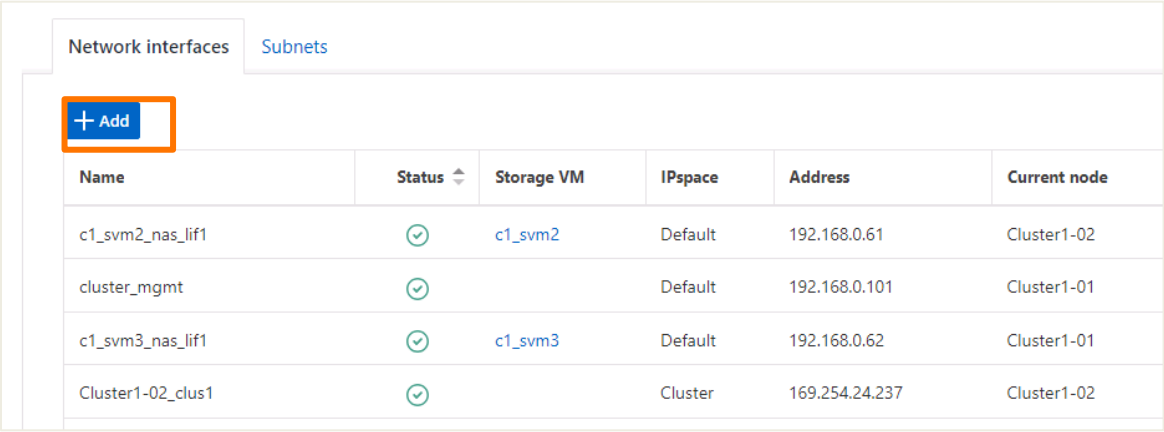
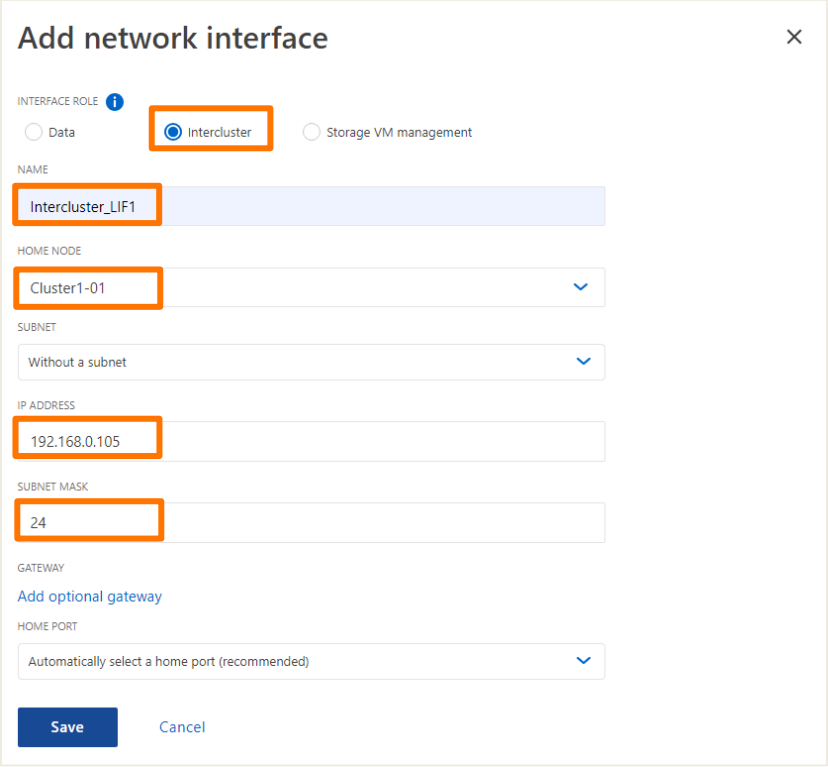
Step	Action
1-3	In the Network interfaces tab, click **Add**. 
1-4	In the Add network interface dialog box, specify the following values: ▪ Interface Role: **Intercluster** ▪ Name: **Intercluster_LIF1** ▪ Home Node: **Cluster1-01** ▪ IP Address: **192.168.0.105** ▪ Subnet Mask: **24** *(automatically generated)* 
1-5	Click **Save**.

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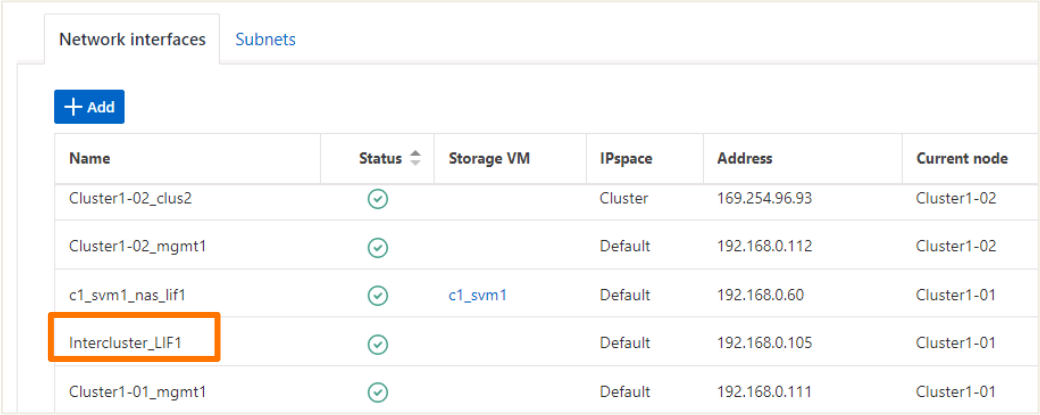
Step	Action
1-6	ONTAP software can automatically select a port for new network interfaces. Therefore, the port selection option is set to **Automatically select a home port (recommended)**. To manually choose a port, you can click and select it from the Home Port drop-down menu. You can move or rehome a port through the UI.
1-7	Verify that Intercluster_LIF1 appears in the Network interfaces list. 
1-8	For fault tolerance, each node of the cluster must have an intercluster LIF. In the next steps, you create an intercluster LIF named Intercluster_LIF2 on node 2.
1-9	From the Network interfaces tab, click **Add**.

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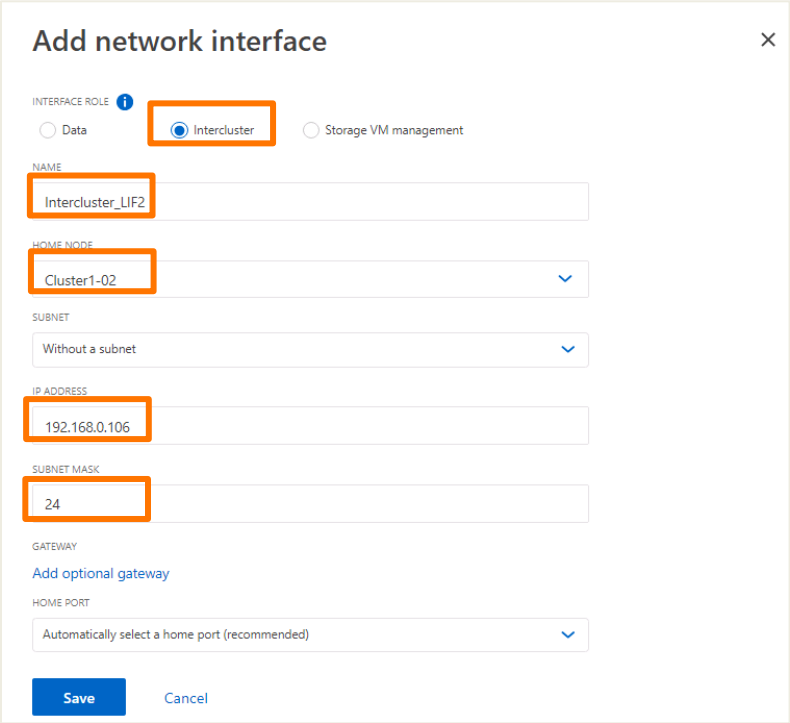
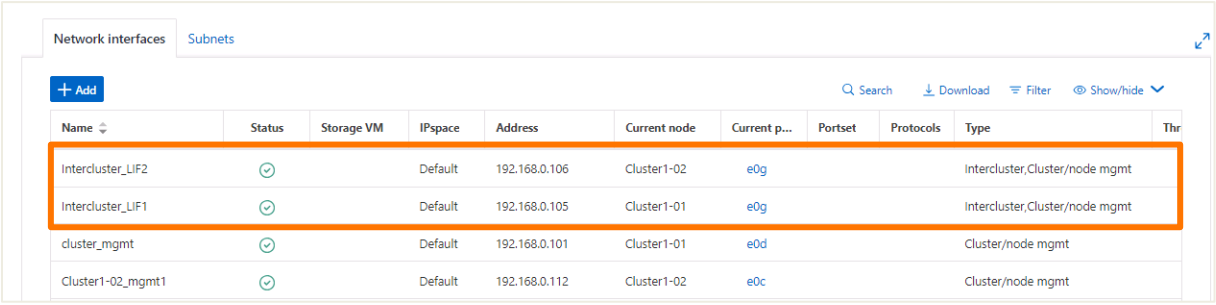
Step	Action
1-10	In the Add network interface dialog box, specify the following values: ▪ Interface Role: **Intercluster** ▪ Name: **Intercluster_LIF2** ▪ Home Node: **Cluster1-02** ▪ IP Address: **192.168.0.106** ▪ Subnet Mask: **24** *(automatically generated)* 
1-11	Click **Save**.
1-12	Verify that the two new intercluster LIFs (Intercluster_LIF1 and Intercluster_LIF2) are listed for Cluster1. 

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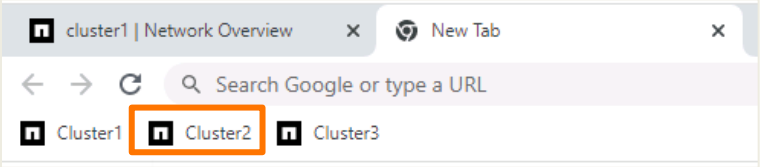
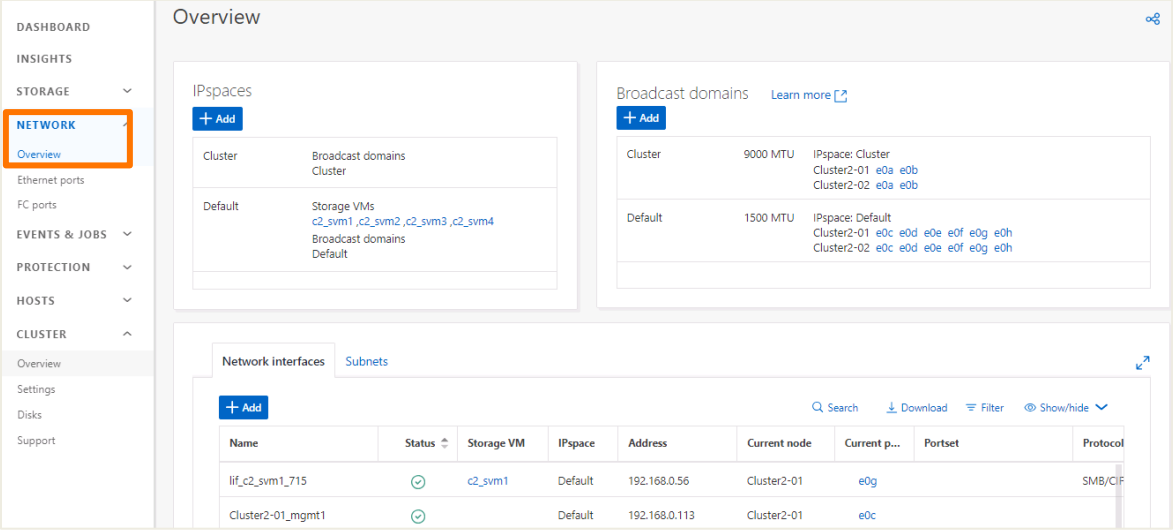
Task 2: Prepare the Storage Environment on Cluster2 as the Secondary Target

In this task, you repeat the steps that you used in Task 1 to create an intercluster LIF on the secondary cluster (Cluster2).

Step	Action
2-1	In your browser, open System Manager for Cluster2 in a new tab, ignore any security warning that might appear, and proceed to log in. 
2-2	Log in to System Manager for Cluster2 by using the following credentials: ▪ User name: **admin** ▪ Password: **Netapp1!**
2-3	In the System Manager navigation pane, select **Network > Overview**. 
2-4	Alternatively, you can create intercluster LIFs from the Protection > Overview pane.

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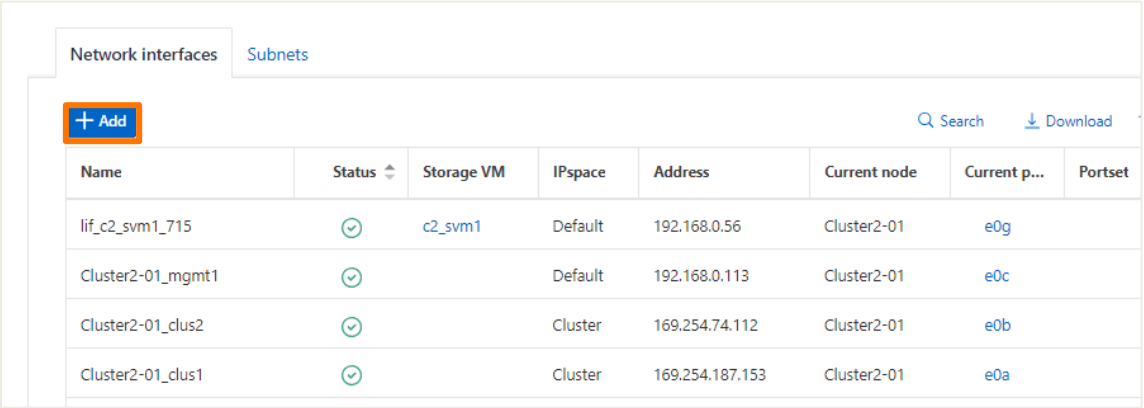
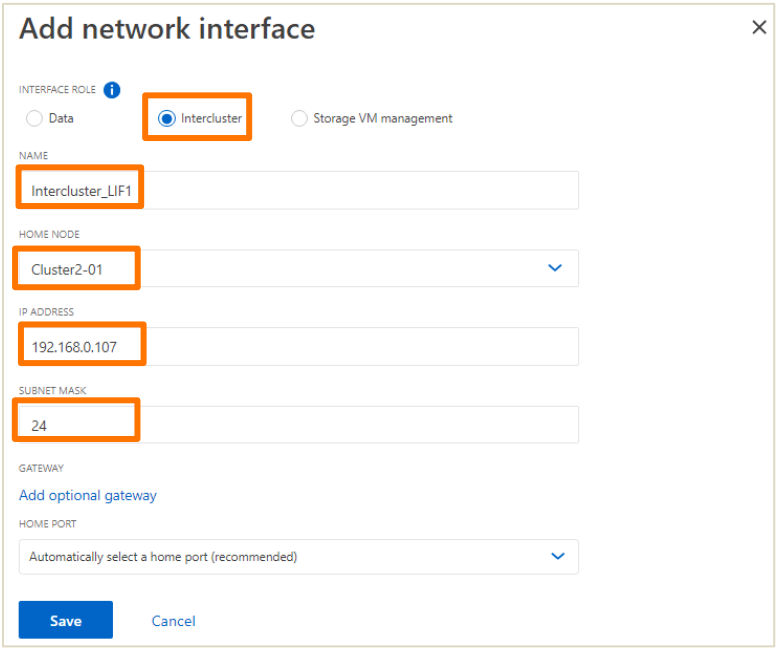
Step	Action
2-5	In the Overview > Network interfaces tab, click **Add**. 
2-6	In the Add network interface dialog box, specify the following values: ▪ Interface Role: **Intercluster** ▪ Name: **Intercluster_LIF1** ▪ Home Node: **Cluster2-01** ▪ IP Address: **192.168.0.107** ▪ Subnet Mask: **24** *(automatically generated)* 
2-7	Click **Save**.

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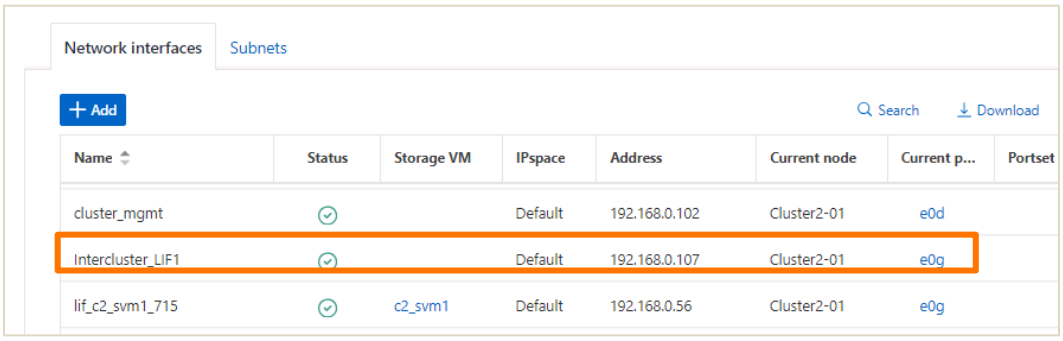
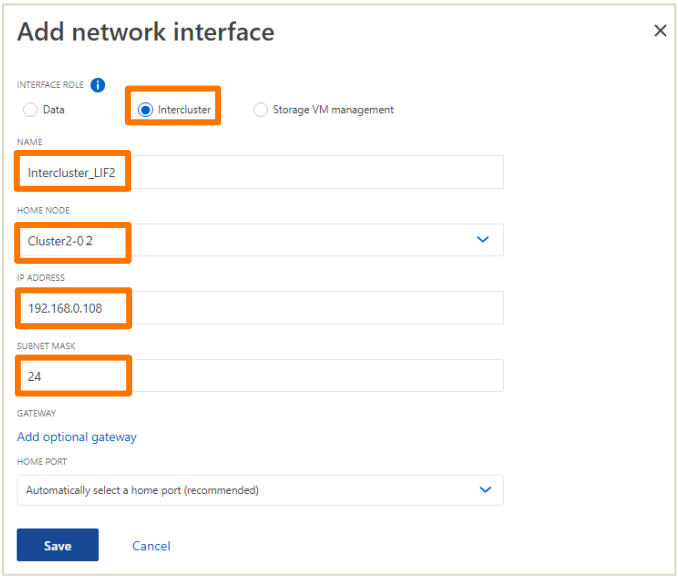
Step	Action
2-8	Verify that Intercluster_LIF1 appears in the Network interfaces list. 
2-9	In the Network interfaces tab, click **Add**.
2-10	In the Add network interface dialog box, specify the following values: ▪ Interface Role: **Intercluster** ▪ Name: **Intercluster_LIF2** ▪ Home Node: **Cluster2-02** ▪ IP Address: **192.168.0.108** ▪ Subnet Mask: **24** *(automatically generated)* 
2-11	Click **Save**.

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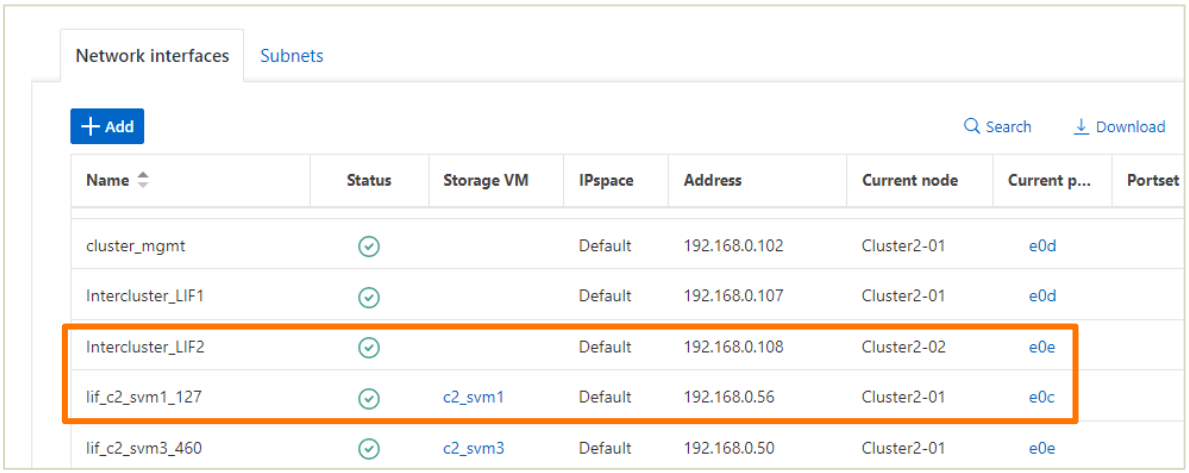
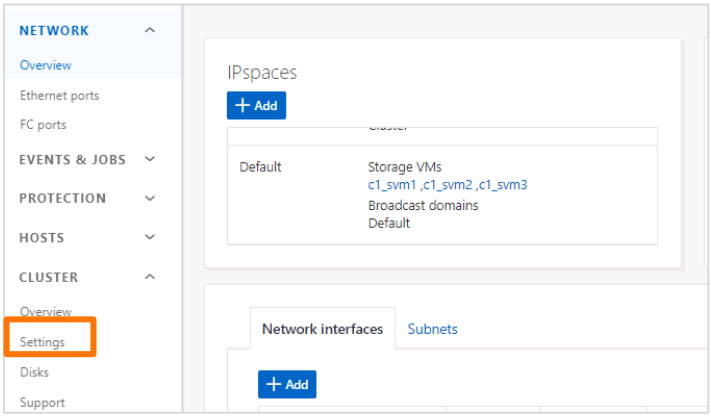
Step	Action
2-12	Verify that the two new intercluster LIFs (Intercluster_LIF1 and Intercluster_LIF2) are listed for Cluster2. 

Task 3: Configure Cluster Peering

Cluster peering is a requirement of intercluster communication. In this task, you configure a peer relationship between Cluster1 and Cluster2. You must create the peer relationship on Cluster1 and then use the same steps to create the relationship on Cluster2.

Step	Action
3-1	On Cluster1, in the System Manager navigation pane, select **Cluster > Settings**. 
3-2	Alternatively, you can create peer clusters from the Protection > Overview pane.
3-3	Perform the following steps: a. Scroll down to the Intercluster settings section. b. In the Intercluster settings > Network interfaces pane, verify the intercluster LIF details for Cluster1. **c.** In the Intercluster settings > Cluster peers pane, click **"Add a peer cluster"**

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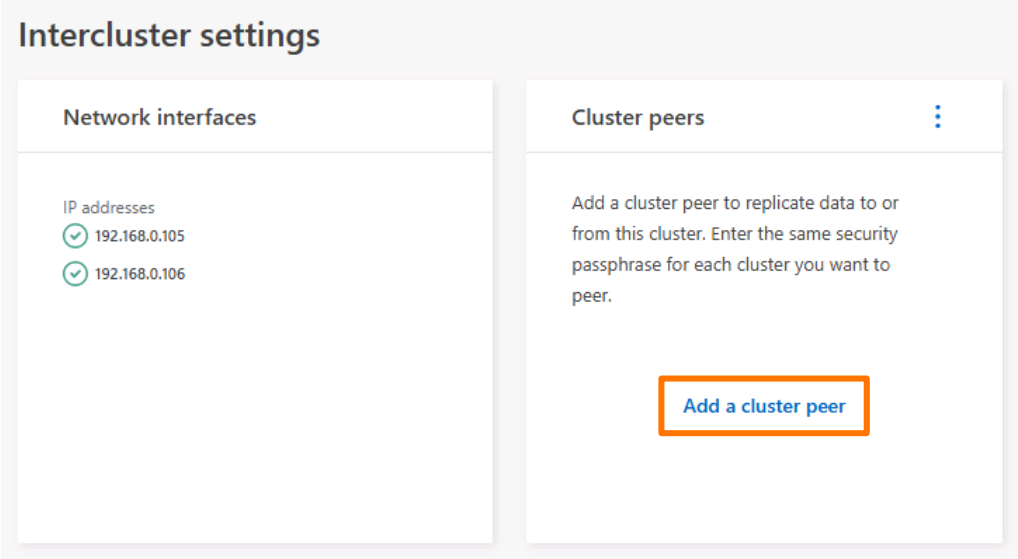
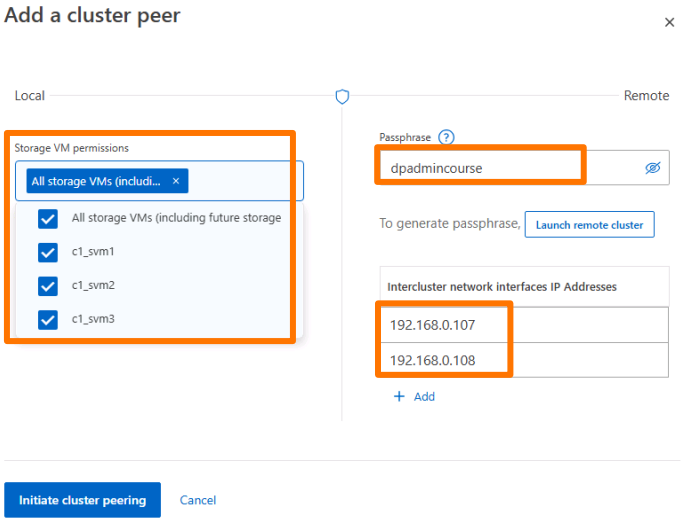
Step	Action
	
3-4	In the Peer cluster dialog box, perform the following steps: a. For Passphrase, enter **dpadmincourse** and then click outside the text box. b. If you want to verify the passphrase, in the Passphrase text box, click the **eyeball icon**. c. In response to the question “Is the relationship encrypted?”, click **Yes**. d. Click **Add**. e. In the Intercluster network interfaces IP Addresses field, enter **192.168.0.107** and **192.168.0.108**. f. Click outside the text box. g. In the Storage VM Permissions field, select the **All storage VMs (including future storage VMs)** checkbox. 

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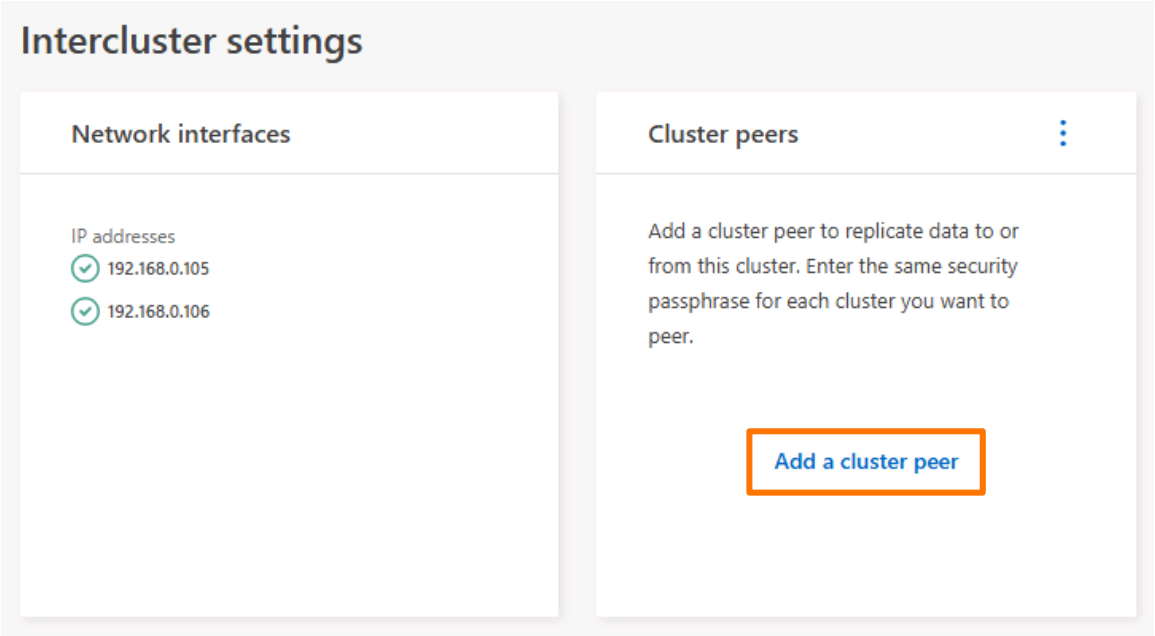
Step	Action
3-5	The passphrase must be at least eight characters. You can use any passphrase, but you must use the same passphrase on both clusters.
3-6	Click **Initiate cluster peering**.
3-7	Perform the following steps for Cluster2: a. In the System Manager navigation pane, select **Cluster > Settings**. b. In the Intercluster settings > Cluster peers pane, click **Add a cluster peer**. 
3-8	In the Peer cluster dialog box, perform the following tasks: a. For Passphrase, enter **dpadmincourse**. b. For relationship encryption, click **Yes**. c. Add two Intercluster network interfaces IP Addresses: **192.168.0.105** and **192.168.0.106**. d. In the Storage VM Permissions field, select the **All storage VMs (including future storage VMs)** checkbox.

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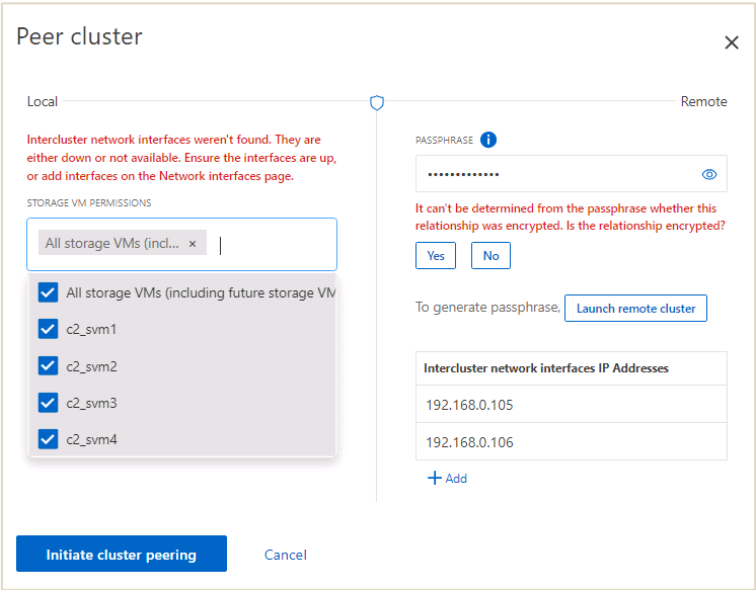
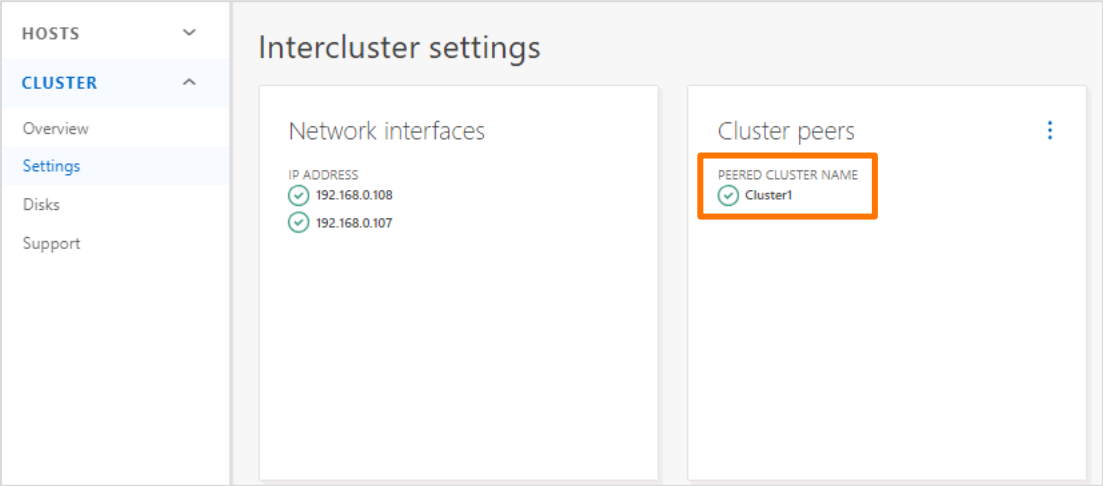
Step	Action
	
3-9	Click **Initiate cluster peering** and wait a few seconds for the peering operation to finish.
3-10	In the Intercluster Settings > Cluster Peers pane, verify that Cluster1 appears, indicating that the cluster peering is successful on Cluster2. 
3-11	On Cluster1, in the System Manager navigation pane, select **Cluster > Settings**.
3-12	In the Intercluster settings > Cluster peers pane, wait for the status icon to turn green to confirm that cluster peering is successful on Cluster1.

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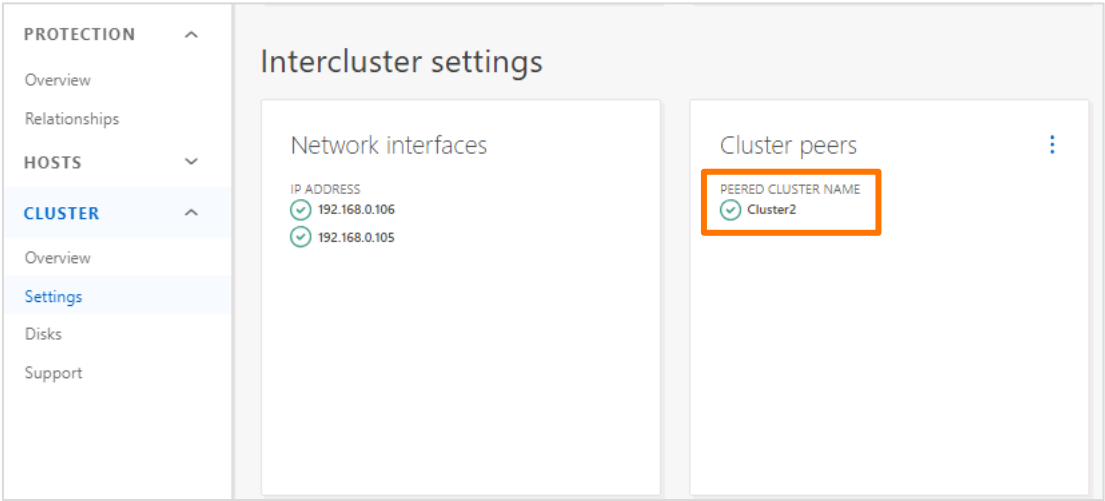
Step	Action
	
3-13	In the CLI for both Cluster1 and Cluster2, run the following commands to review the cluster peering status: • `cluster peer show` • `cluster peer health show` • `cluster peer ping`

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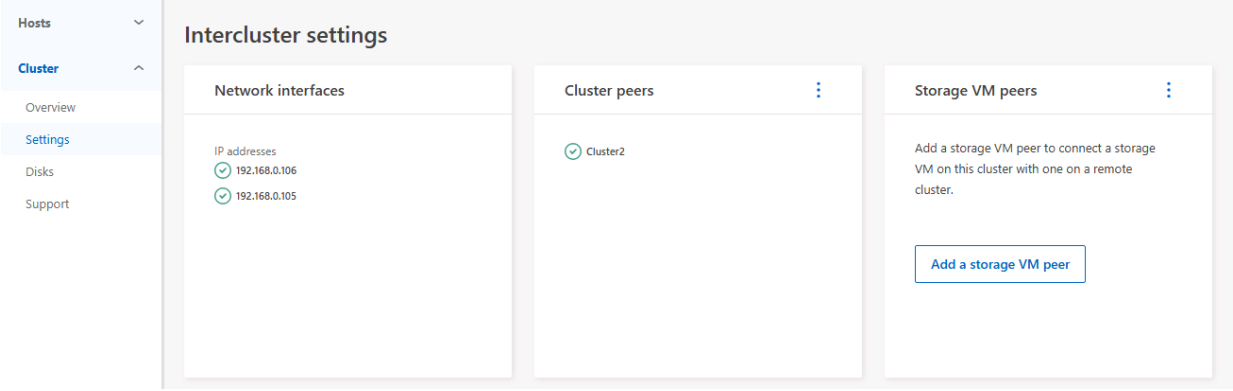
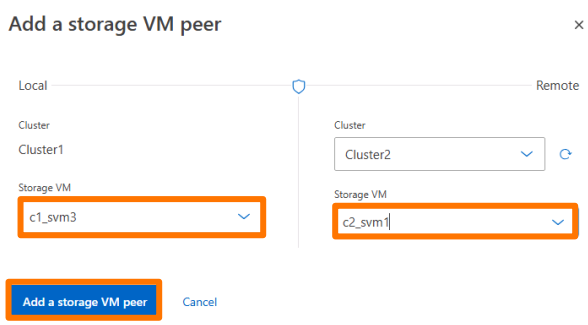
Task 4: Configure Storage VM Peering

In this task, you create a peer relationship between c1_svm3 on Cluster1 and c2_svm1 on Cluster2.

Step	Action
4-1	Perform the following steps for Cluster1: a. In the System Manager navigation pane, select **Cluster > Settings**. b. In the Intercluster settings > Storage VM peers pane, click the **three vertical dots**, and then select **Peer storage VMs**. 
4-2	In the Peer storage VMs dialog box, provide the following information: ▪ Cluster1 Storage VM: **c1_svm3** ▪ Cluster2 Storage VM: **c2_svm1** 
4-3	Click **Add a storage VM peer**

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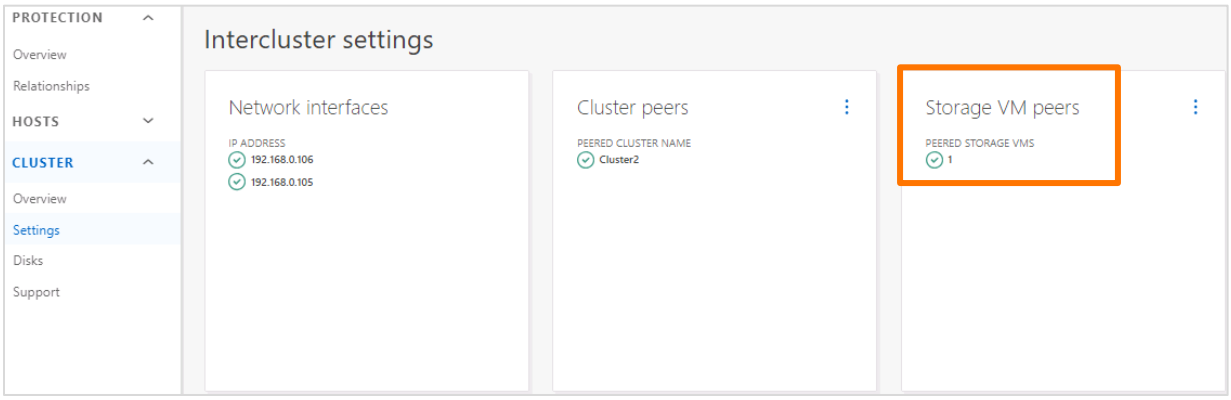
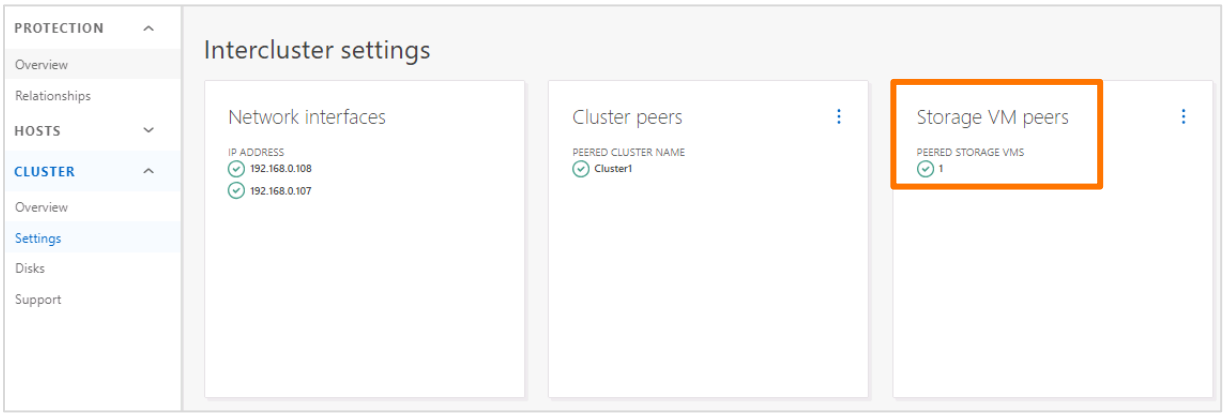
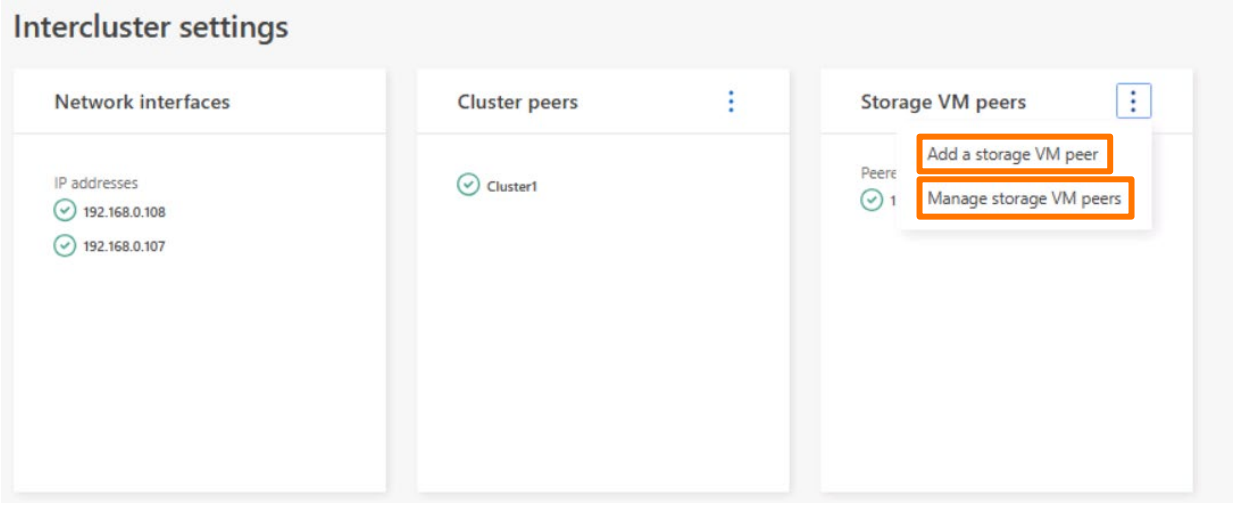
Step	Action
4-4	Verify that the peering is successful and is listed in the Storage VM peers pane. 
4-5	For Cluster2, in the System Manager navigation pane, select **Cluster > Settings**.
4-6	In the Intercluster settings > Storage VM peers pane, verify that the peering is successful and is listed. 
4-7	It can take a few seconds for the status to change.
4-8	In the Intercluster settings > Storage VM peers pane, click the **three vertical dots**, and then select **Manage storage VM peers**. 

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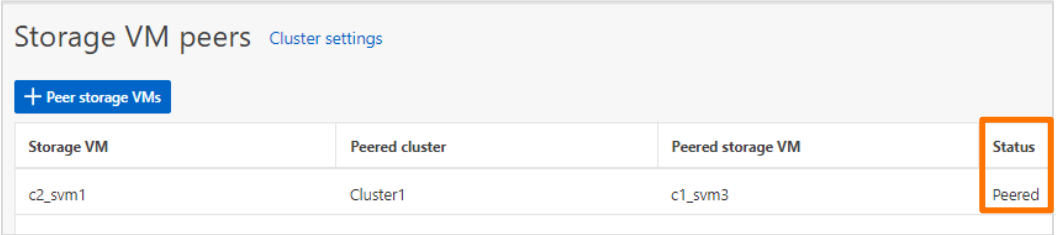
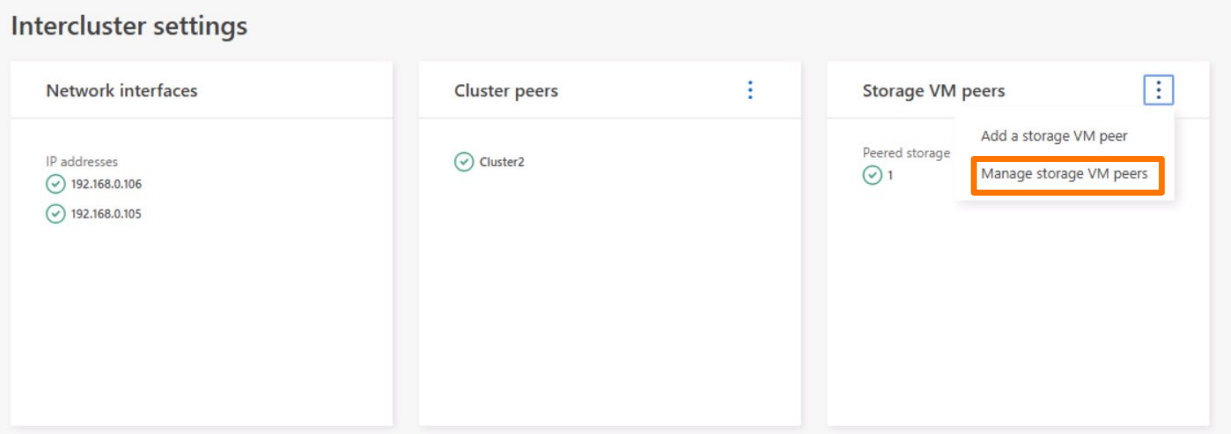
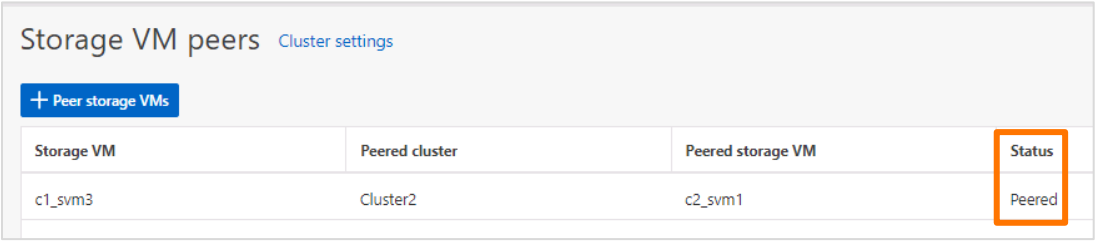
Step	Action
4-9	On the Storage VM peers page, verify that the storage VM peer relationship is listed and that the status is Peered. 
4-10	In Cluster 1, in the System Manager navigation pane, select **Cluster > Settings**.
4-11	In the Intercluster Settings > Storage VM peers pane, click the **three vertical dots**, and then select **Manage storage VM peers**. 
4-12	On the Storage VM peers page, verify that the storage VM peer relationship is listed and that the status is Peered. 

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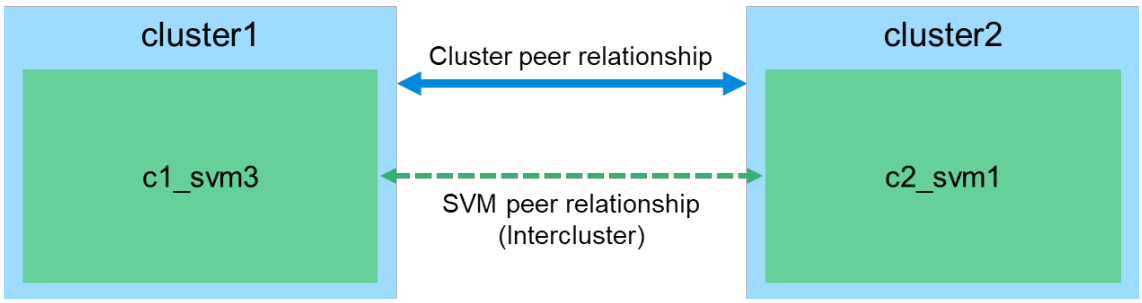
Task 5: Review the Exercise Environment

In this task, you review the exercise environment that you created in the previous tasks.

Step	Action
5-1	Verify that the following actions occurred: ▪ You configured intercluster LIFs on each node of the two ONTAP clusters. ▪ You established peer relationships between the Cluster1 and Cluster2 ONTAP clusters. ▪ You created a peer relationship between storage VMs c1_svm3 and c2_svm1. 

End of exercise

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